

Career / Job Application Tracking System

Long-Term Backend + Data Engineering Learning Roadmap

Goal of This Project

Build ONE continuously evolving project that teaches:

- Python fundamentals
- PostgreSQL + SQL deeply
- Backend engineering concepts
- Data engineering workflows
- APIs
- ETL
- Analytics/reporting
- Deployment basics
- Real project architecture evolution

The project is intentionally designed to grow gradually instead of building many disconnected tutorial projects.

CURRENT PROJECT DOMAIN

Core Idea

A relational Job / Career Application Tracking System.

Entities:

- Students
- Companies
- Jobs
- Applications

This domain is excellent because it naturally supports:

- CRUD
- relationships
- joins

- analytics
 - reporting
 - workflows
 - validations
 - search/filtering
 - ETL later
 - APIs later
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PHASE 1 — PYTHON + SQL FOUNDATIONS (CURRENT)

Main Focus

Learn backend fundamentals through CLI + PostgreSQL.

Concepts Covered

Python

- functions
- loops
- nested logic
- validation
- modular files
- exception handling
- reusable utilities
- control flow
- formatting

SQL

- CREATE TABLE
- INSERT
- SELECT
- UPDATE
- DELETE

- WHERE
- ORDER BY
- LIMIT
- aggregates
- GROUP BY
- subqueries
- joins
- foreign keys
- constraints

Database Design

- relational modeling
- one-to-many relationships
- many-to-many relationships
- junction tables
- referential integrity

Backend Thinking

- validation layers
- business rules
- workflow logic
- reusable helpers
- query organization
- user flow
- CLI UX

Project Features

Students Module

- CRUD
- search
- sorting

- aggregations

Companies Module

- CRUD
- search
- grouping
- analytics

Jobs Module

- CRUD
- salary filtering
- sorting
- company relationships

Applications Module

- relational workflows
- date handling
- status tracking
- foreign key validation

Relational Reports

- joins
- reporting queries
- analytics-style SQL

Skills Gained

After this phase you should be comfortable with:

- writing SQL manually
- thinking relationally
- building CRUD systems
- debugging backend logic
- handling user validation
- structuring small backend projects

Resume Value After Phase 1

Suitable for:

- beginner backend internships
 - Werkstudent applications
 - GitHub showcase
 - technical discussions
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PHASE 2 — REFACTORING + PROFESSIONALIZATION

Main Goal

Improve architecture and scalability.

Planned Improvements

Validation Improvements

- reusable existence validators
- controlled status values
- cancel/back workflow handling
- better error messages

Architecture Improvements

- split menus into functions
- reduce repetition
- improve abstraction quality
- cleaner workflow orchestration

UX Improvements

- pagination
- searching before selection
- partial updates
- cleaner display formatting

Backend Concepts Learned

- abstraction design

- maintainability
- refactoring
- separation of concerns
- scalable workflow design

Skills Gained

You start thinking less like:

"How do I make this work?"

and more like:

"How should systems be designed and maintained?"

PHASE 3 — ADVANCED SQL + ANALYTICS

Main Goal

Turn project into analytics/reporting system.

Concepts to Add

SQL

- advanced joins
- CASE statements
- window functions
- CTEs
- views
- indexes
- query optimization

Reporting Features

- top companies
- hiring trends
- application success rates
- salary analytics
- country-level analytics

Data Thinking

- reporting pipelines
- analytical queries
- dashboard-style data

Skills Gained

This phase starts building:

- data analyst thinking
- data engineering foundations
- reporting/query optimization intuition

PHASE 4 — PANDAS + DATA ENGINEERING WORKFLOWS

Main Goal

Introduce ETL and data workflows.

Features

CSV Import/Export

- export applications
- import bulk jobs
- bulk student uploads

Pandas

- dataframe transformations
- cleaning data
- aggregations
- reporting
- preprocessing

ETL Workflows

- extract from database
- transform with pandas
- load processed data

Automation Concepts

- scheduled reports
- data pipelines
- reusable ETL scripts

Skills Gained

This phase builds:

- practical data engineering skills
- pandas experience
- ETL understanding
- pipeline thinking

This becomes very useful for:

- data engineering internships
 - backend/data hybrid roles
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PHASE 5 — FASTAPI BACKEND

Main Goal

Convert CLI system into real backend API.

Features

REST APIs

- GET endpoints
- POST endpoints
- PUT/PATCH endpoints
- DELETE endpoints

Backend Concepts

- request validation
- JSON responses
- status codes
- API architecture
- routers

- schemas/models

Pydantic

- validation models
- serialization
- type safety

Database Integration

- API ↔ PostgreSQL connection
- query handling
- response formatting

Skills Gained

This phase creates real backend development experience.

Now project becomes:

- portfolio-worthy backend system
- deployable service foundation
- realistic API project

PHASE 6 — DEPLOYMENT + DEVOPS BASICS

Main Goal

Learn how real systems are configured and deployed.

Concepts

Docker

- containerization
- Dockerfile
- docker-compose

Environment Management

- .env files
- configuration handling
- secrets management basics

Logging

- structured logs
- debugging practices

Deployment

- Render/Railway/Fly.io/etc.
- PostgreSQL hosting
- API deployment

Skills Gained

This phase introduces:

- deployment workflows
 - backend operations basics
 - environment configuration
 - production-style thinking
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PHASE 7 — OPTIONAL ADVANCED EXTENSIONS

Possible Future Additions

Authentication

- login/signup
- JWT
- role-based access

Async Processing

- async FastAPI
- background tasks

Search Systems

- advanced filtering
- pagination
- full-text search

Testing

- pytest
- API testing
- integration tests

Cloud Concepts

- AWS/GCP basics
 - object storage
 - managed databases
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WHAT THIS PROJECT TEACHES OVERALL

Backend Engineering

- relational systems
- APIs
- workflows
- architecture
- validation
- business logic
- scalability

Data Engineering

- ETL
- reporting
- analytics
- pipelines
- pandas workflows
- SQL depth

Software Engineering

- refactoring
- maintainability
- modularity

- debugging
- project evolution

Professional Skills

- Git workflow
 - iterative development
 - long-term project ownership
 - technical communication
 - architectural thinking
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WHY THIS APPROACH IS STRONG

Instead of:

- many shallow tutorial projects
- disconnected tech demos
- copy-paste CRUD apps

You build:

- one evolving realistic system
- continuous engineering growth
- increasing architectural depth
- real debugging experience
- scalable learning path

This mirrors how real software systems evolve.

TARGET OUTCOME BEFORE GERMANY

A realistic and achievable target within your timeline:

Must Complete

- Phase 1 fully
- most of Phase 2
- at least introductory Phase 4 or 5

Ideal Outcome

- CLI relational system complete
- strong joins/reporting
- pandas workflows started
- FastAPI basics added
- GitHub repo polished

At that stage you would already have:

- meaningful backend portfolio project
- strong SQL foundation
- growing data engineering exposure
- realistic project discussion material for interviews
- long-term project that can continue evolving even after reaching Germany